



INSIGHTSMASTERY ACADEMY

POWER BI ROW-LEVEL SECURITY STUDENT HANDBOOK

COMPLETE EDITION

From Absolute Beginner to Advanced

A Complete Guided Learning Path for Securing Power BI Data

ROLES • RULES • MANAGE ROLES • VIEW AS • TEST AS ROLE
STATIC RLS • DYNAMIC RLS • USERPRINCIPALNAME • CUSTOMDATA
SECURITY TABLES • BRIDGE TABLES • BIDIRECTIONAL SECURITY FILTERS
WORKSPACE ROLES • READ / BUILD / WRITE • EMBEDDED IDENTITY
OBJECT-LEVEL SECURITY • TROUBLESHOOTING • MINI PROJECT

Professional Training Edition

Designed for Classroom & Self-Learning

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1. How to Use This Handbook

This handbook teaches **row-level security (RLS)** in Power BI from the very first idea to production-grade designs: dynamic security, bridge tables, managers who see their whole team, embedded applications and object-level security. Every chapter uses the same small **Sales** table, so you can check every number with a calculator. Facts were checked against Microsoft Learn on 3 October 2026; anything that was in preview is labelled **in preview at the time of writing**.

1.1 The learning path

Stage	Chapters	You will be able to...
1. Foundations	Chapter 3 – Chapter 4	Explain roles, rules and how a rule filters a model.
2. Build and test	Chapter 5 – Chapter 7	Create, test and publish roles; know which users are filtered.
3. Dynamic security	Chapter 8 – Chapter 9	Filter by the signed-in user with security and bridge tables.
4. Patterns	Chapter 10	Apply 16 tested patterns, from one line to a manager hierarchy.
5. Real-world edges	Chapter 11 – Chapter 14	Handle embedding, storage modes, OLS, alternatives, debugging.
6. Prove it	Chapter 15 – Chapter 18	Build a mini project, solve exercises, keep a one-page reference.

Teaching pattern. Each important feature is explained in the same order: **WHAT** → **WHY** → **SYNTAX / WHERE TO CLICK** → **EVERY WORD** → **SIMPLE EXAMPLE** → **RESULT** → **WHY RESULT** → **REAL-WORLD EXAMPLE** → **COMMON MISTAKE** → **PRACTICE**.

1.2 Legend and how to study

You see	It means
Result: North sales = 131,000	A checked result. Every number is computed from the sample data.
«Show only the rows where Region is North.»	Read it like English — the rule in plain words.
Modeling → Manage roles	A click path in Power BI Desktop or the Power BI service.
[Region] = "North"	DAX code, exactly as you type it.
On / Off	A setting that is switched on or off.
  	Blue box: analogy, tip or rule · Amber box: common mistake · Green box: good practice or practice task.

How to study: read a chapter once, then open Power BI Desktop, type the sample data (**Chapter 2**) and repeat every example. If your numbers differ from the green **Result** lines, your rule or your relationships differ — find out why before moving on. Finish with the exercises and the mini project.



2. Sample Data and Practice File

All IMA handbooks share one 10-row table called **Sales**. You will meet the same numbers in the DAX, modeling and visualization courses, which makes it easy to spot when a security rule changes a total.

2.1 The Sales table

ID	Date	Customer	Product	Category	Region	Qty	Unit Price	Sales Amount	Cost	Sales-person
1	2026-01-01	Amit	Laptop	Electronics	North	2	50,000	100,000	80,000	Ravi
2	2026-01-02	Neha	Mouse	Electronics	South	5	1,000	5,000	3,000	Priya
3	2026-01-03	Rahul	Keyboard	Electronics	North	3	2,000	6,000	3,500	Ravi
4	2026-01-04	Amit	Monitor	Electronics	West	2	15,000	30,000	22,000	Amit
5	2026-01-05	Priya	Chair	Furniture	South	4	5,000	20,000	14,000	Priya
6	2026-01-06	Rahul	Desk	Furniture	North	1	15,000	15,000	10,000	Ravi
7	2026-01-07	Neha	Laptop	Electronics	South	1	50,000	50,000	40,000	Priya
8	2026-01-08	Amit	Chair	Furniture	West	2	5,000	10,000	7,000	Amit
9	2026-01-09	Rahul	Mouse	Electronics	North	10	1,000	10,000	6,000	Ravi
10	2026-01-10	Priya	Desk	Furniture	South	2	15,000	30,000	20,000	Priya

Type the table into Power BI Desktop with **Home** → **Enter data**, name it **Sales**, and set **Date** to the Date type and the four number columns to Whole number.

2.2 The four measures used everywhere

Measures:

```
Total Sales      = SUM ( Sales[SalesAmount] )
Total Cost       = SUM ( Sales[Cost] )
Profit           = [Total Sales] - [Total Cost]
Total Quantity   = SUM ( Sales[Quantity] )
```

Read it like English:

«Add up SalesAmount for the rows I am allowed to see. Do the same for Cost and Quantity. Profit is sales minus cost.»

i Why measures matter for RLS

Measures are evaluated after security has removed rows. So the same measure, `Total Sales`, shows 276,000 to a user who sees every row and 131,000 to a user who sees only North. You never write security logic inside your measures.



2.3 Numbers to recognise

Numbers to recognise

Slice	Rows (SalesID)	Total Sales	Total Cost	Profit	Quantity
All rows	1,2,3,4,5,6,7,8,9,10	276,000	205,500	70,500	32
Region = North	1,3,6,9	131,000	99,500	31,500	16
Region = South	2,5,7,10	105,000	77,000	28,000	12
Region = West	4,8	40,000	29,000	11,000	4
Category = Electronics	1,2,3,4,7,9	201,000	154,500	46,500	23
Category = Furniture	5,6,8,10	75,000	51,000	24,000	9
Salesperson = Ravi	1,3,6,9	131,000	99,500	31,500	16
Salesperson = Priya	2,5,7,10	105,000	77,000	28,000	12
Salesperson = Amit	4,8	40,000	29,000	11,000	4

i Notice

Each salesperson works in exactly one region: Ravi = North, Priya = South, Amit = West. That is why a region rule and a salesperson rule give the same totals in this data set. In real data they usually differ — always check which column your rule really uses.

2.4 Illustrative security tables

RLS needs to know *who* may see *what*. The standard Sales table has no user information, so this handbook adds small **illustrative** tables. They are invented for teaching; the e-mail domain **ima-demo.com** is fictitious.

Users — one row per person who opens the report

UserEmail	DisplayName	SeeAll
ravi@ima-demo.com	Ravi	FALSE
priya@ima-demo.com	Priya	FALSE
amit@ima-demo.com	Amit	FALSE
meera@ima-demo.com	Meera	FALSE
kiran@ima-demo.com	Kiran	TRUE



UserRegion — which regions each person may see (a bridge table)

UserEmail	Region
ravi@ima-demo.com	North
priya@ima-demo.com	South
amit@ima-demo.com	West
meera@ima-demo.com	North
meera@ima-demo.com	South

Meera is a sales manager for two regions, so she has two rows. Kiran is the director: instead of listing every region, the **SeeAll** flag in Users says she may see everything.

Employee — salespeople and the reporting hierarchy

EmployeeID	Name	ManagerID	Email	Path = PATH(EmployeeID, ManagerID)
1	Kiran	(blank)	kiran@ima-demo.com	1
2	Meera	1	meera@ima-demo.com	1 2
3	Ravi	2	ravi@ima-demo.com	1 2 3
4	Priya	2	priya@ima-demo.com	1 2 4
5	Amit	1	amit@ima-demo.com	1 5

Employee has one row per person, so **Name** is unique and can filter Sales[Salesperson]. Kiran manages Meera and Amit; Meera manages Ravi and Priya. Kiran and Meera sell nothing themselves, so they have no Sales rows of their own. The **Path** column is a calculated column used by the hierarchy pattern in [Chapter 10](#).

RegionDim — one row per region

A tiny dimension table with one column, **Region**, and three rows: North, South, West. Rules work best on small dimension tables like this one (see [Chapter 9](#)).



2.5 The practice model

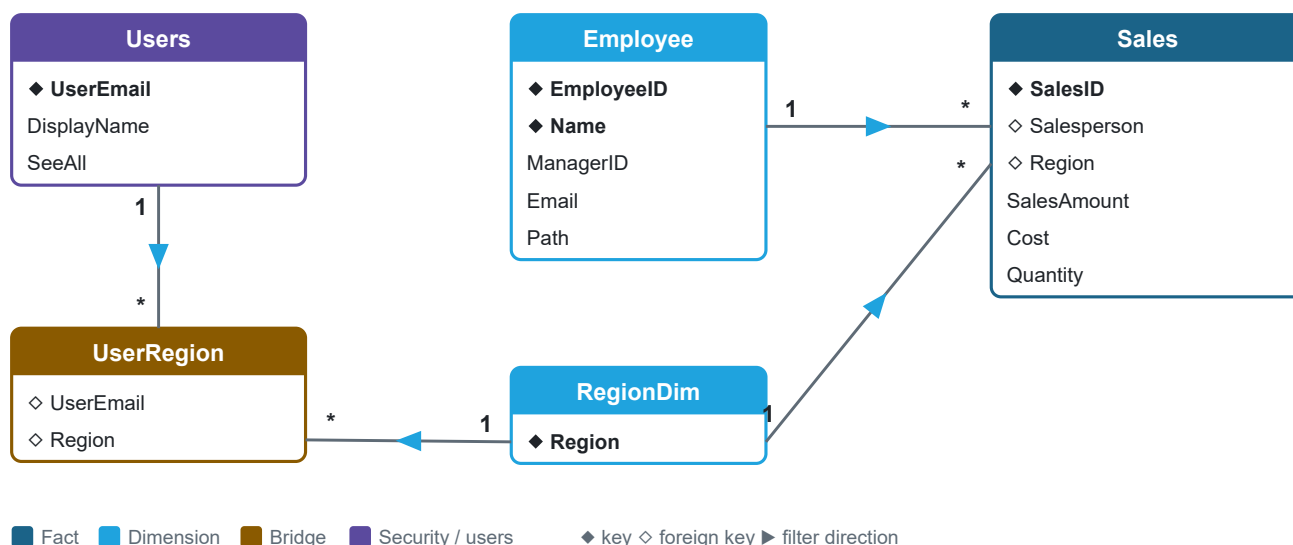


Figure 2.1 — The practice model: two dimensions filter Sales; Users and UserRegion hold the security mapping

Relationship (one → many)	Used by
Employee[Name] → Sales[Salesperson]	Dynamic salesperson rule and the manager hierarchy
RegionDim[Region] → Sales[Region]	Every region rule
RegionDim[Region] → UserRegion[Region]	Bridge patterns (one user, many regions)
Users[UserEmail] → UserRegion[UserEmail]	Bridge patterns and the SeeAll flag

All relationships start as one-to-many with a single filter direction. **Chapter 9** explains which way each filter flows and when you need **Apply security filter in both directions**.

	#1 North	#2 South	#3 North	#4 West	#5 South	#6 North	#7 South	#8 West	#9 North	#10 South
No role (admin)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Region = North	✓		✓			✓			✓	
Region = South		✓			✓		✓			✓
Region = West				✓				✓		

Figure 2.2 — Which Sales rows survive each simple region rule (cyan = visible)



3. What Is Row-Level Security?

Row-level security (RLS) lets one semantic model show different rows to different people. A **semantic model** is the set of tables, relationships and measures behind a report. RLS works with **filters** that hide rows, and those filters live inside **roles**. By default a model has no roles at all.

3.1 Why RLS exists

Without RLS you would need one report per audience: one for North, one for South, one for West. With RLS you publish one model and one report. Each viewer opens the same page and sees only the rows a role allows. The most common design is **dynamic RLS**: a single role that filters per user through a table that maps each user to the data they may see.

i Analogy — one building, many keycards

Think of the model as an office building and the report as the lift. Everybody uses the same lift, but each keycard opens only the floors it was issued for. A role is a type of keycard; the rule printed on it decides which floors (rows) open. Someone without any keycard reaches no floor at all.

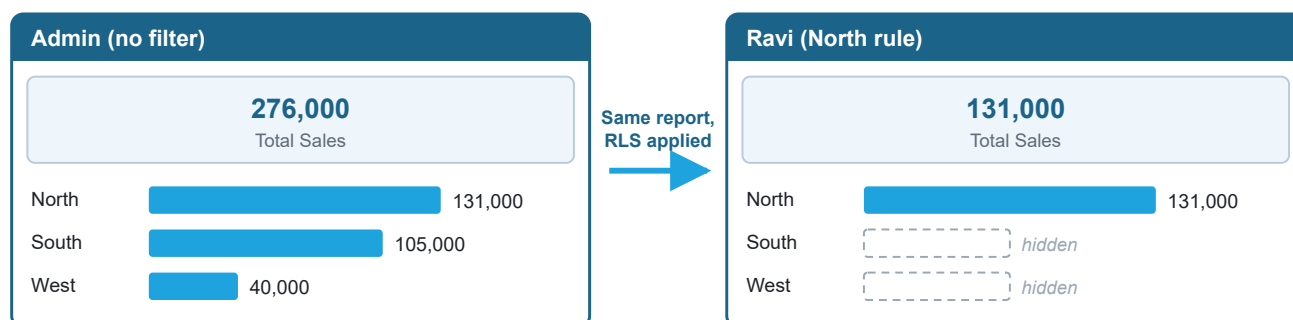


Figure 3.1 — The same report page before and after a North rule

The admin sees all 10 rows and 276,000 in sales. Ravi's role keeps only the 4 North rows, so his card shows 131,000 and the South and West bars simply do not exist for him.

3.2 Vocabulary

Term	Meaning in this handbook
Semantic model	Tables, relationships and measures that reports query.
Role	A named container for security rules. Each role has a unique name and usually one or more rules.
Rule (filter expression)	A DAX expression on one table that returns TRUE or FALSE for each row. Only TRUE rows stay visible.
Member	A user or group assigned to a role in the Power BI service.
Static rule	Uses a fixed value, for example <code>[Region] = "North"</code> .
Dynamic rule	Uses a function that returns who is signed in, such as <code>USERPRINCIPALNAME()</code> , <code>USERNAME()</code> or <code>CUSTOMDATA()</code> .



Term	Meaning in this handbook
UPN	User principal name — the sign-in identifier, for example ravi@ima-demo.com. It is not always the same as the email address.
Effective identity	The identity a query runs as. In embedded apps it is passed in the embed token (Chapter 11).
Viewer	The workspace role whose users are filtered by RLS.
Read / Build / Write	Permissions on a semantic model. Read and Read+Build are filtered; Write is not.
OLS	Object-level security: hides whole tables or columns instead of rows (Chapter 12).

3.3 RLS, OLS and workspace permissions

Three different controls are often confused. RLS filters **rows only**: if a user can see a row, they see every column of that row. RLS also cannot show a summary while hiding the detail behind it — a user sees details and totals of the same allowed rows, nothing more.

Question	RLS	OLS	Workspace permissions
What does it protect?	Rows of a table	Whole tables or columns	Access to items in the workspace
Hide one column?	No	Yes	No
Show a total but hide its detail rows?	No	No	No
Applies to Admin, Member, Contributor?	No	No	Defines who those people are
Applies to Viewers?	Yes	Yes	Defines who those people are

⚠ Common mistake

Expecting RLS to hide a column such as Cost. RLS never removes columns. Use OLS for that, or remove the column from the model.



4. How RLS Works Inside the Model

This chapter opens the box. You do not need it to click through the menus, but it explains every surprising result you will meet later.

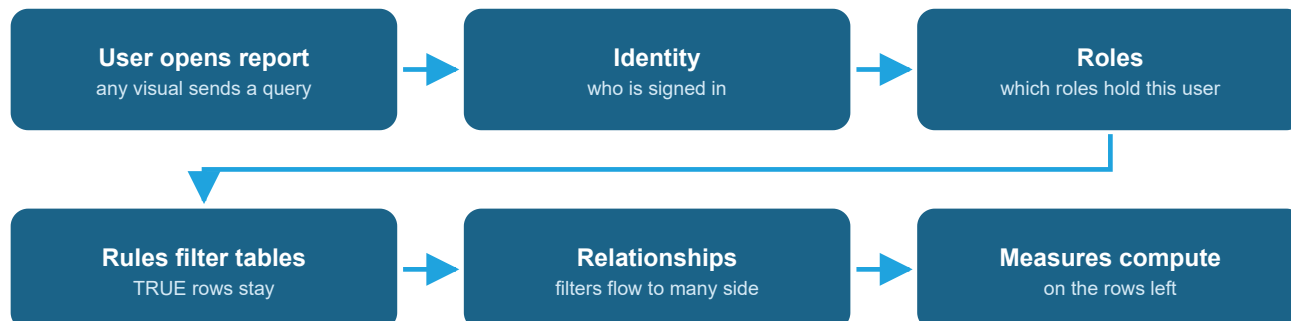


Figure 4.1 — The path of every query when RLS is on

4.1 A rule is a TRUE/FALSE test per row

A rule is evaluated in **row context**: Power BI takes the table the rule belongs to, looks at one row at a time and asks the rule "TRUE or FALSE?". Rows that answer TRUE stay; every other row is removed. `FALSE()` returns FALSE for every row and so blocks the whole table; `TRUE()` keeps every row.

Rule on RegionDim:

```
[Region] = "North"
```

Read it like English:

«For each RegionDim row, keep it only if its Region is North.»

Result: RegionDim keeps 1 row (North); Sales keeps rows 1,3,6,9; Total Sales = 131,000.

4.2 Filters travel along relationships

A rule on a table also filters the related rows **in the many direction**. RegionDim is on the one side of RegionDim[Region] → Sales[Region], so the North rule on RegionDim removes every Sales row that is not North. Security follows only **active** relationships; an inactive relationship never carries the filter. Filtering a small dimension table like RegionDim is generally more efficient than filtering the fact table directly.

Security filters move in **one direction** by default — from the one side to the many side — even when the relationship itself is set to filter both ways. **Chapter 9** shows when and how to change that with **Apply security filter in both directions**.

A table that has **no rule** is fully readable, unless a filter reaches it through a relationship. In our model Employee has no rule in the North role, so all employees remain listed; only their Sales rows outside North disappear.

4.3 Inside one role: AND. Across roles: OR

When one role has rules on two tables, a row must pass **both** — the filters are intersected. When a user belongs to **two roles**, the roles are **additive**: the user sees every row allowed by either role. "Once denied, always denied" does not apply: a user in a `FALSE()` role and a `TRUE()` role sees all rows.



	#1 North	#2 South	#3 North	#4 West	#5 South	#6 North	#7 South	#8 West	#9 North	#10 South
Role A: Region = North	✓		✓			✓			✓	
Role B: Category = Electronics	✓	✓	✓	✓			✓		✓	
One role with both rules (AND)	✓		✓						✓	
Member of A and B (OR)	✓	✓	✓	✓		✓	✓		✓	

Figure 4.2 — One role with two rules intersects; two roles add up

Setup	Visible Sales IDs	Rows	Total Sales
One role: North AND Electronics	1,3,9	3	116,000
Two roles: North OR Electronics	1,2,3,4,6,7,9	7	216,000
Two roles: FALSE() and TRUE()	1,2,3,4,5,6,7,8,9,10	10	276,000

Result: Intersection gives 116,000; union gives 216,000. Prefer one role that describes everything a user needs.

4.4 Measures run after security

RLS removes rows first; then every measure is computed on what is left. That is why `[Total Sales]` for Ravi is 131,000, not 276,000. A DAX expression cannot switch RLS off and cannot even detect that RLS is active. A "% of all regions" measure therefore needs an unsecured summary table ([Section 10.15](#)).

Once a model has roles, a Viewer who is **not a member of any role** sees no data at all. Every visual shows (Blank) — no rows.

⚠ Common mistake

Writing a measure with `ALL()` to "escape" the filter. `ALL()` only removes report filters; rows hidden by RLS are not in the model for that user, so they never come back.



5. Creating Roles in Power BI Desktop

Roles and rules are written in Power BI Desktop with DAX, published with the model, and filled with members later in the Power BI service.

5.1 Steps: create the role 'North Sales'

1. Select **Modeling** → **Manage roles**.
2. Select **New** and type the role name **North Sales**. Role names can't contain a comma — `London, ParisRole` is rejected.
3. Under **Select tables**, choose **RegionDim**.
4. Under **Filter data**, build the rule Region equals North (default editor), or select **Switch to DAX editor** and type the expression below.
5. In the DAX editor, select the **checkmark** to validate the expression (the **X** reverts it).
6. Select **Save**.

Rule on RegionDim (DAX editor):

```
[Region] = "North"
```

Read it like English:

«Keep a RegionDim row only when its Region is North.»

Result: Sales rows 1,3,6,9 remain; Total Sales = 131,000, Total Cost = 99,500, Profit = 31,500.

5.2 Default editor or DAX editor

What it is:

The role screen has two editors. The **default editor** builds a rule from drop-down lists. The **DAX editor** is a text box where you type any DAX rule, with IntelliSense.

Why it matters:

Simple static rules are quicker in the default editor. Dynamic rules that use `USERNAME()` or `USERPRINCIPALNAME()` can only be written in the DAX editor.

Syntax / where to click:

Switch to DAX editor / **Switch to default editor** on the role screen.

Every word explained:

Switch to DAX editor — shows the current rule as DAX text.

Checkmark — validates the expression.

x — reverts your unsaved change.

save — stores all roles in the model.



Simple example:

```
[Region] IN { "North", "West" }
```

Result: Sales rows 1,3,4,6,8,9; Total Sales = 171,000.

Why this result:

IN tests whether the region is in the list, so North and West rows return TRUE.

Real-world example:

A regional director role covering two regions is one rule, not two roles.

⚠ Common mistake

Switching back to the default editor after typing DAX it cannot display. Power BI warns that information may be lost — select **Cancel** to keep your DAX. Also: type commas as argument separators, even if your Windows region uses semicolons.

✓ Practice

Create a role 'Electronics' on Sales with `[Category] = "Electronics"`. Expected Total Sales: 201,000.

Editor capabilities

Capability	Default editor	DAX editor
Static equal / not-equal rule	On	On
Any DAX expression (IN, AND, dates)	Off	On
USERNAME() / USERPRINCIPALNAME()	Off	On
IntelliSense	Off	On
Checkmark validation / X revert	Off	On

5.3 What Desktop cannot do

- **Assign users.** Members are added only in the service (**Chapter 7**).
- **Live connections to Analysis Services.** Roles can't be defined in Desktop; define them in the Analysis Services model instead.

5.4 Alternative: define roles on the web

When you edit a semantic model in the Power BI service, the same role screen is available:

1. Open the model for editing and select **Manage roles** on the ribbon.
2. Select **New**, then **Select tables** and **Filter data**; select **Save**.
3. Select **Assign**, add people or groups, and select **Save**.

Expected behaviour: The role appears in the model and, after Assign, already has members.



6. Testing Roles: View as and Test as Role

Never publish a role you have not tested. You test twice: in Desktop before publishing and in the service after adding members.

6.1 In Desktop: View as

What it is:

Modeling → **View as** shows the report as a member of the role you choose.

Why it matters:

It proves the rule and relationships give the right rows before anybody else sees the report.

Syntax / where to click:

1. Select **Modeling** → **View as**.
2. Select a role.
3. Optional: tick **Other user** and type a UPN, such as ravi@ima-demo.com.
4. Select **OK**.

Every word explained:

Role — the role whose rules are applied.

Other user — the identity returned by the identity functions. It changes results only for dynamic rules, and you must tick the role as well.

Simple example:

View as role 'North Sales'.

Result: Total Sales = 131,000 (rows 1,3,6,9).

Why this result:

The North rule keeps the North RegionDim row, which filters Sales.

⚠ Common mistake

Ticking only Other user without the role. No role is applied, so nothing is tested.

✓ Practice

View as Other user = priya@ima-demo.com with the dynamic role from **Chapter 8**. Expected: 105,000.

6.2 In the service: Test as role

1. Open the model's **Security** page (**Chapter 7**).
2. Select **More options (...)** next to the role → **Test as role**.
3. Use **Now viewing as** to switch to another role, a combination of roles, or a specific person.
4. Use **Viewing** to open another report in the same workspace.
5. Select **Back to Row-Level Security** when finished.

**i Your identity is used**

Test as role runs dynamic rules with **your** UPN. If you are not in the Employee table, a rule like Employee[Email] = USERPRINCIPALNAME() shows (Blank) — no rows. Choose a specific person to test someone else, and check guests and embedded apps with real accounts or tokens.

6.3 A test plan with expected totals

Test	Who / roles	Expected Total Sales
Static North	Role North Sales	131,000
Combination	Roles North Sales + West (union)	171,000
Dynamic, Ravi	Dynamic role + Other user ravi (Chapter 8)	131,000
Dynamic, Amit	Dynamic role + Other user amit	40,000
Dynamic, Meera	Dynamic role + Other user meera (no Sales as salesperson)	(Blank) — no rows
Unknown user	Dynamic role + Other user not in Employee	(Blank) — no rows
Block	Role with FALSE ()	(Blank) — no rows

Always include an **unexpected** value, such as an unknown user. A safe rule returns no rows for it.

6.4 A 'Who Am I' measure

```
Who Am I =
-- show both identity functions on a card while testing
USERNAME () & " | " & USERPRINCIPALNAME ()
```

Expected behaviour: In Desktop the card shows DOMAIN\User | your UPN; in the service both parts show your UPN.

6.5 What cannot be tested

- View as and Test as role don't work for DirectQuery models with single sign-on (SSO) enabled.
- Dashboards can't be tested; you are redirected to the report.
- Q&A visuals, Quick insights and Copilot can't be validated; paginated reports aren't supported.

6.6 Measure the performance cost

RLS adds filters to every query. In Desktop open **Performance Analyzer**, record the visuals once without a role, then again with **View as**, and compare the durations.

Expected behaviour: Two sets of query durations for the same visuals; the difference between them is the cost of the RLS filters.

⚠ Common mistake

Testing only as yourself. As a workspace Admin you are not filtered, so the report looks correct for you and wrong for everyone else.



7. Publishing, Members and Workspace Permissions

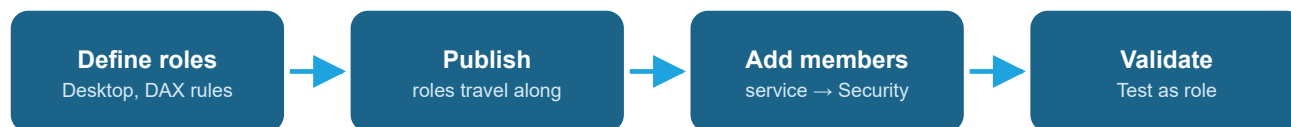


Figure 7.1 — The RLS workflow from Desktop to the service

7.1 Add members to a role

1. Publish the report from Desktop to a workspace.
2. In the workspace, open **More options (...)** on the semantic model → **Security**.
3. On the Row-Level Security page, select a role.
4. Type the email or name of a user or group and add it.

Expected behaviour: The number in parentheses next to the role name shows how many members it has. Remove a member with its **X**. The page works only when the model already has roles.

To assign members you need write access to the semantic model (for example workspace Admin, Member or Contributor); model ownership or Build may also be required — check with your workspace admin.

Supported members	Not supported
Individual users, external users, distribution groups, mail-enabled groups, Microsoft Entra security groups	Microsoft 365 groups; groups created in Power BI

Good practice: map security groups to roles whenever possible, so joiners and leavers are handled in the group, not in every model.

7.2 Who is filtered?

Only **Viewer**-level access is filtered; workspace Admins, Members and Contributors are not.

Permission on the model	Typical source	RLS enforced
Read	Viewer role, shared report, app	Yes
Read + Build	Viewer with Build (e.g. Analyze in Excel)	Yes
Write	Admin / Member / Contributor, or per-item Write	No

To enforce RLS, give report consumers the Viewer role, or share the report or publish an app to them. Making them workspace Members gives them Write: they see all 10 rows (276,000). Read/Build users in no role see (Blank) — no rows.

7.3 Limits to remember

Slicer and filter selections live outside the model, so RLS doesn't protect them. A visual using an object hidden by OLS shows the deleted-field message (**Chapter 12**). **Publish to web** is not available for reports that rely on RLS.



8. Dynamic RLS and Identity Functions

A **static** rule contains a fixed value, such as `[Region] = "North"`. A **dynamic** rule asks Power BI *who is signed in* and filters with that answer. Microsoft describes the dynamic approach as the most common one: a single role, filtered per user through a table that maps users to the data they may see.

8.1 Why dynamic beats one role per region

	Static: one role per region	Dynamic: one role for everybody
Roles in the model	North, South, West ... one per audience	One, e.g. SalesUser
Rule	<code>[Region] = "North"</code> (a constant)	<code>Employee[Email] = USERPRINCIPALNAME ()</code>
New salesperson joins	Edit the model, add or change a role, publish again	Add one row to a table; refresh

Dynamic rules use `USERNAME ()`, `USERPRINCIPALNAME ()` or `CUSTOMDATA ()`.

8.2 The identity functions

USERPRINCIPALNAME()

What it is:

A DAX function with no arguments. It returns the **user principal name (UPN)** of the person who connected — the name they sign in with, for example `ravi@ima-demo.com`.

Why it matters:

It is the standard way to write a dynamic rule: compare a column in your security table with the signed-in user's UPN.

Syntax / where to click:

```
USERPRINCIPALNAME ()
```

Type it in the DAX editor of a role: **Modeling** → **Manage roles** → select the role → select the table → **Switch to DAX editor**. Dynamic rules can only be written in the DAX editor.

Every word explained:

USERPRINCIPALNAME — Name of the function: “give me the user principal name”.

() — Empty brackets — the function takes no parameters.

Returns — Text: the UPN at the moment the connection is made.

Simple example:

```
-- Rule on the Employee table
Employee[Email] = USERPRINCIPALNAME ()
```

Result: Ravi signs in → Total Sales = 131,000 (rows 1,3,6,9, 4 rows).



Why this result:

The rule keeps the one Employee row whose Email equals Ravi's UPN. That row filters Sales through the relationship Employee[Name] → Sales[Salesperson].

⚠ Common mistake

Storing the user's *e-mail address* instead of the UPN. They are often equal, but not always — see “UPN versus e-mail” below.

✓ Practice

Write the rule above, then use the Who Am I card from [Chapter 6](#) to see what USERPRINCIPALNAME() returns for you in Desktop.

USERNAME()

No arguments; returns the domain and user name of the connection. In Desktop that is `DOMAIN\User`; in the service it is the UPN. Avoid mixing it with USERPRINCIPALNAME() in one model.

```
Who Am I =  
USERNAME ()
```

Result: Desktop card: DOMAIN\User. Service card: ravi@ima-demo.com.

USEROBJECTID()

Returns the user's **Microsoft Entra Object ID** (the SID for SQL Server Analysis Services). Use it when your security table stores IDs, never compare an ID column with a UPN.

```
-- Rule on a table that stores object IDs  
Users[ObjectID] = USEROBJECTID ()
```

Result: Only rows mapped to the user's ID are visible; an unknown ID gives (Blank) — no rows.

CUSTOMDATA()

Returns the **CustomData** property of the connection string, blank if not set. Mainly used by embedded apps via `EffectiveIdentity.CustomData` ([Section 10.14](#)). Not supported in DirectQuery mode in calculated columns or RLS rules.

```
-- Rule on RegionDim; the app passes the region  
RegionDim[Region] = CUSTOMDATA ()
```

Result: No CustomData passed → blank never equals a region → (Blank) — no rows.



8.3 Comparison of identity functions

Function	Returns	In Desktop	In the service	Typical use
<code>USERPRINCIPALNAME ()</code>	UPN at connection time	<code>user@contoso.com</code>	UPN	Standard dynamic RLS
<code>USERNAME ()</code>	Domain and user name	<code>DOMAIN\User</code>	UPN	Older models; avoid mixing
<code>USEROBJECTID ()</code>	Entra Object ID (SID in SSAS)	Object ID / SID	Object ID	Tables that store IDs
<code>CUSTOMDATA ()</code>	CustomData of the connection; blank if not set	Blank unless set	Value set by the app	Embedded apps (EffectiveIdentity.CustomData)

8.4 Getting the comparison right

UPN versus e-mail. The UPN is the sign-in identifier, not necessarily the e-mail address. Store exactly what `USERPRINCIPALNAME()` returns, not the Entra **mail** attribute (in this handbook both are `ravi@ima-demo.com` on purpose).

Case. DAX compares text without caring about case, so `RAVI@IMA-DEMO.COM` stored in `Employee[Email]` still matches.

Result: Total Sales = 131,000 (rows 1,3,6,9, 4 rows) — identical to the lower-case result.

Guests (B2B). `USERPRINCIPALNAME()` may return `user@partner.com` or `user_partner.com#EXT#@tenant.onmicrosoft.com`; the format is not guaranteed, so validate with a real guest. Groups containing guests might not evaluate correctly: add guests directly by e-mail or use dynamic RLS.

8.5 Your first dynamic rule

Goal: each salesperson sees only his or her own sales, using one role.

1. Open **Modeling** → **Manage roles** and select **New**. Name the role **SalesUser**.
2. Under **Select tables**, choose **Employee**.
3. Select **Switch to DAX editor** and type the rule below.
4. Select the checkmark to validate, then **Save**.
5. Test with **Modeling** → **View as**: tick **SalesUser** and **Other user**, type the UPN, OK.

Rule on Employee:

```
-- Keep only the Employee row of the signed-in user
Employee[Email] = USERPRINCIPALNAME ()
```

Read it like English:

«Keep the Employee row of the signed-in person — and, through the relationship, only their sales.»



Signed-in user	USERPRINCIPALNAME()	Sales rows	Total Sales	Profit
Ravi	ravi@ima-demo.com	1,3,6,9	131,000	31,500
Priya	priya@ima-demo.com	2,5,7,10	105,000	28,000
Amit	amit@ima-demo.com	4,8	40,000	11,000
Meera	meera@ima-demo.com	—	(Blank) — no rows	(Blank)
stranger (not in Employee)	stranger@ima-demo.com	—	(Blank) — no rows	(Blank)

Expected behaviour: Ravi, Priya and Amit each see only their own rows. Meera is in Employee but sells nothing herself, so she sees (Blank) — no rows. An unknown user sees (Blank) — no rows.

Meera is a manager. To let her see her team you need either a bridge table ([Chapter 9](#)) or the hierarchy pattern ([Section 10.13](#)).

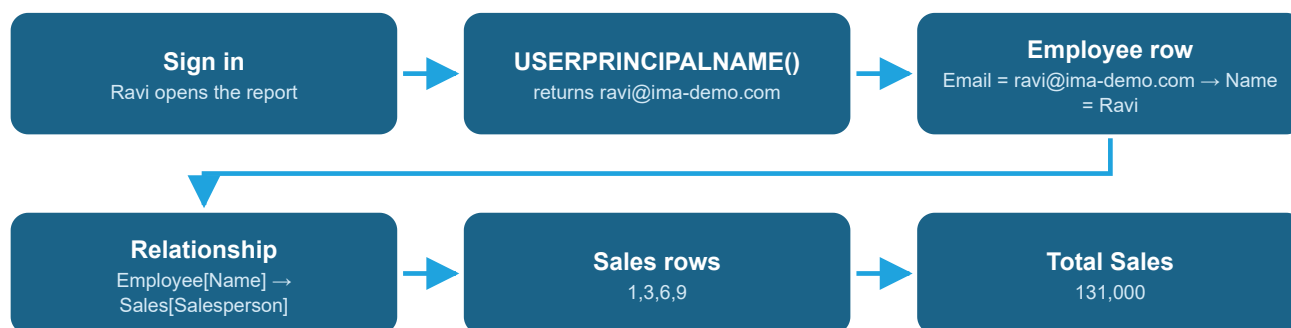


Figure 8.1 — How the signed-in identity becomes a filtered total

⚠ Common mistake

Testing with **View as** → **Other user** but without ticking the role. In the service, **Test as role** uses **your** identity; check it with the Who Am I card ([Chapter 6](#)).

8.6 Safe defaults

Test with expected *and* unexpected values, and write rules so an unexpected value returns **no rows** — as the rule above does for unknown users.

Pattern on Sales (illustrative; the practice Users table has no UserType column):

```

-- Safe default: unknown user types see nothing
VAR UserType = LOOKUPVALUE ( Users[UserType], Users[UserEmail], USERPRINCIPALNAME () )
VAR MyName = LOOKUPVALUE ( Employee[Name], Employee[Email], USERPRINCIPALNAME () )
RETURN
    IF ( UserType = "Manager", TRUE (),
        IF ( UserType = "Worker", [Salesperson] = MyName, FALSE () ) )
  
```

Expected behaviour: Manager → all rows; Worker → own rows; anything else → (Blank) — no rows.



9. Security Tables, Relationships and Bidirectional Filters

A rule filters one table. Everything else depends on how that filter travels through the relationships of your model. This chapter shows how to shape the model so the filter reaches Sales.

9.1 Filter dimensions, not facts

In a **star schema** small **dimension** tables (RegionDim, Employee) describe the large **fact** table (Sales). Filtering a dimension is generally more efficient than filtering the fact table. A row filter applies to the rows of its table **and to related rows in the many direction**.

Example: `RegionDim[Region] = "North"` (3-row dimension) is preferred over `Sales[Region] = "North"`; both give Total Sales = 131,000.

9.2 Only active relationships carry security

RLS filters travel only through **active** relationships (solid lines). An inactive relationship (dashed line) carries nothing. If two tables have several relationships, security follows the active one.

Caution: with RLS enabled, `USERRELATIONSHIP()` in queries or measures might cause unexpected errors.

9.3 LOOKUPVALUE or a relationship?

`LOOKUPVALUE` can be used inside a rule to fetch the user's attribute — for example his region — by matching the identity against a login column:

```
-- Rule on RegionDim: fetch the user's region
RegionDim[Region] =
    LOOKUPVALUE ( UserRegion[Region], UserRegion[UserEmail], USERPRINCIPALNAME () )
```

Result: Ravi → Total Sales = 131,000 (rows 1,3,6,9, 4 rows).

This works for users with one region. Microsoft's guidance is to **avoid LOOKUPVALUE when relationships can do the job**. A user like Meera, with two UserRegion rows, needs a rule that handles several values — see [Section 10.11](#).

9.4 The bridge problem

Meera may see North and South: two rows in the **bridge** table UserRegion. The rule sits on Users: `Users[UserEmail] = USERPRINCIPALNAME ()`.

Follow the filter: Users → UserRegion goes from the one side to the many side, so it flows. But RegionDim → UserRegion points *into* the bridge. To reach RegionDim — and from there Sales — the filter must travel from the many side (UserRegion) up to the one side (RegionDim). By default **security filters are single-direction, even when a relationship is bidirectional**.

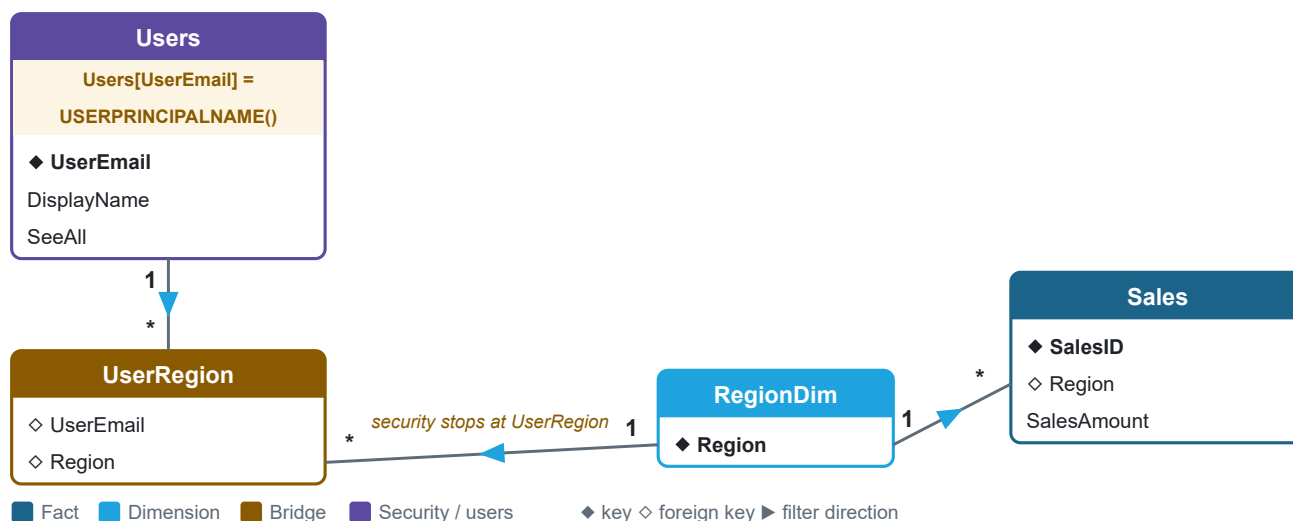


Figure 9.1 — Single direction: the security filter reaches UserRegion and stops; RegionDim and Sales are not filtered

⚠ Security leak

In this model the rule on Users removes every Users row except Meera's, and UserRegion keeps her two rows. Nothing filters RegionDim, so nothing filters Sales. Meera sees **every row: Total Sales = 276,000** (rows 1,2,3,4,5,6,7,8,9,10). It is not (Blank); it is **all data**. Every other user in the role sees the same.

Fix: apply the security filter in both directions

1. Open the relationship between **RegionDim** and **UserRegion** in the **Edit relationship** dialog.
2. Set **Cross filter direction** to **Both**.
3. Tick **Apply security filter in both directions**.
4. Select **OK**, then test with **Modeling** → **View as** (role + Other user `meera@ima-demo.com`).

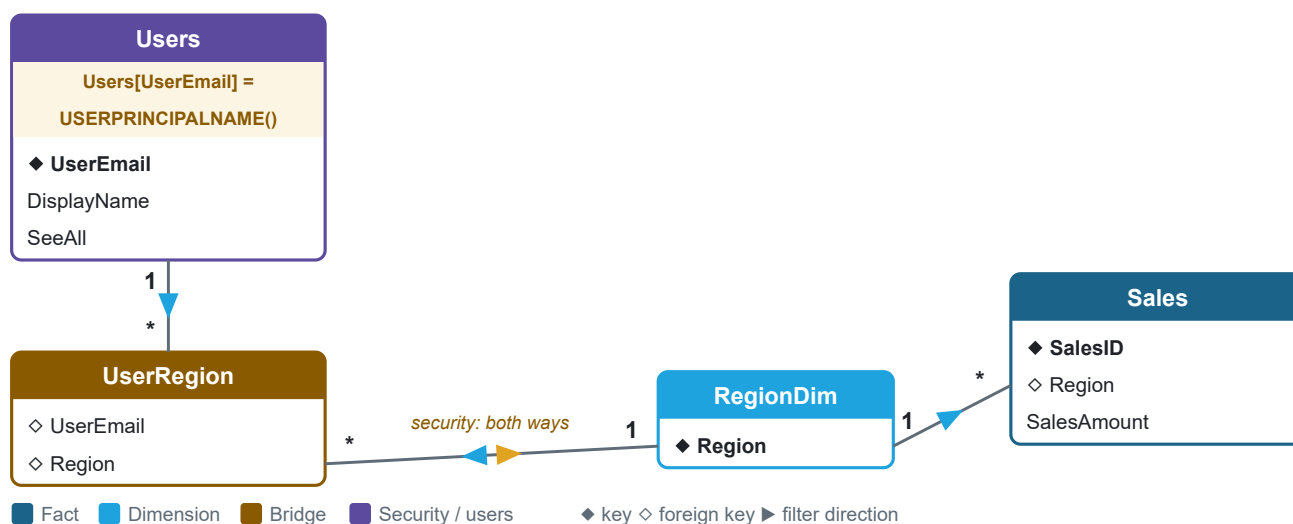


Figure 9.2 — Both directions with security: Users → UserRegion → RegionDim → Sales (amber arrow = security filter going up)

Result: Meera → Total Sales = 236,000 (rows 1,2,3,5,6,7,9,10, 8 rows). Ravi → 131,000; Priya → 105,000; Amit → 40,000; Kiran (no UserRegion rows) → (Blank) — no rows.



⚠ Limits and cost

- If a table takes part in several bidirectional relationships, **Apply security filter in both directions** can be selected for only one of them.
- Bidirectional security filtering can hurt performance. Measure it: compare durations with Performance Analyzer without RLS and with **View as**.

9.5 Relationship settings at a glance

Setting	Value	When to use for RLS
Cross filter direction	Single	Default. Dimension → fact, e.g. RegionDim → Sales.
Cross filter direction	Both	Required before the security option below can be ticked.
Apply security filter in both directions	On	Bridge tables: security must climb from the many side (UserRegion) to the one side (RegionDim).
Apply security filter in both directions	Off	Default. Security flows only one → many, even on a Both relationship.
Relationship is active (solid line)	Yes	Security only travels through active relationships.
Relationship is inactive (dashed line)	No	The relationship carries no security filter at all.

9.6 An alternative without bidirectional filtering

You can get Meera's 236,000 without any both-direction relationship by putting a DAX rule directly on RegionDim that looks up the user's regions in UserRegion. The step-by-step version is [Section 10.11](#); the bidirectional version is [Section 10.10](#).

9.7 Storage-mode notes

- **DirectQuery**: RLS adds filters to every query sent to the source — optimise the source (indexes, persisted computed columns).
- **Composite models**: you can't define RLS on tables from a remote source, local RLS doesn't filter remote tables, and RLS isn't imported from the source. If the Users table is DirectQuery in a composite model, new users appear only after the next refresh.
- Details: [Chapter 11](#).

✓ Common mistake and practice

Forgetting **Apply security filter in both directions** on a bridge model: the rule validates but nobody is restricted. Untick the option, test as Priya, then tick it again. *Answer: all rows without it, 105,000 with it.*



10. RLS Patterns — Step by Step

This chapter is a recipe book. Each section solves one security need with the practice model from [Chapter 2](#). Every recipe was checked against the 10-row Sales table, so you can rebuild it in Power BI Desktop and compare your numbers with the green result lines.

Each pattern gives the **Why**, the setup (table, editor, role), the DAX rule, a **Read it like English** line, any model settings, the **Result** (visible row IDs, Total Sales, Profit) and the most common mistake. The full click path is shown once, in the first pattern.

Test every role with **Modeling** → **View as** in Desktop or **Test as role** in the service ([Chapter 6](#)); members are assigned only in the service ([Chapter 7](#)).

10.1 Static rule: one region

Why: The North sales team should see only North sales. A fixed value (a constant) in the rule is enough.

Steps:

1. **Modeling** → **Manage roles** → **New**, and name the role **North** (role names can't contain a comma).
2. **Select tables** → **RegionDim** → **Filter data**.
3. Pick the column and value in the default editor, or select **Switch to DAX editor** and type the rule below.
4. Select the **checkmark** to validate, then **Save**.

```
[Region] = "North"
```

Read it like English: «Keep a RegionDim row only when its Region is North; Sales follows through the relationship.»

Result: 4 rows (IDs 1,3,6,9) — **Total Sales 131,000**, Profit 31,500

⚠ Common mistake

Putting the rule on **Sales[Region]** as well. The RegionDim filter already reaches Sales through the active one-to-many relationship, so the extra rule is redundant; filtering the smaller dimension table is generally more efficient than filtering the fact table.

10.2 Static rule: a list of values with IN

Why: A regional manager covers two regions. **IN** tests a value against a list, which is shorter and safer than chaining several **||** (OR) tests.

Table: RegionDim · **Editor:** DAX · **Role:** NorthWest. **Rule:**

```
RegionDim[Region] IN { "North", "West" }
```

Read it like English: «Keep the RegionDim rows whose Region is in the list North, West.»

Result: 6 rows (IDs 1,3,4,6,8,9) — **Total Sales 171,000**, Profit 42,500

**⚠ Common mistake**

Writing `RegionDim[Region] = "North" && RegionDim[Region] = "West"`. One row cannot have two regions at once, so the rule is FALSE for every row and the role shows **(Blank) — no rows**. Use `IN` or `||`.

10.3 Static rule on a fact column

Why: The Electronics buyers should see only Electronics sales. The practice model has no Category dimension, so the rule goes directly on the Sales column.

Table: Sales · **Editor:** DAX · **Role:** Electronics. **Rule:**

```
[Category] = "Electronics"
```

Read it like English: «Keep a Sales row only when its Category is Electronics.»

Result: 6 rows (IDs 1,2,3,4,7,9) — **Total Sales 201,000**, Profit 46,500

Filtering dimension tables is generally more efficient than filtering fact tables. A rule on Sales is acceptable here only because there is no Category table; in a larger model, add one and put the rule there.

⚠ Common mistake

Expecting the rule to hide the **Category** column. RLS removes rows, never columns: the role still sees every column of the rows it keeps. To hide columns use OLS ([Chapter 12](#)).

10.4 Two rules in one role = intersection (AND)

Why: A North Electronics specialist should see only rows that are North **and** Electronics. Inside one role, filters on different tables are combined with AND.

Table: RegionDim · **Editor:** DAX · **Role:** NorthElectronics. Still in the same role, select **Sales**, add the category rule, and save once. **Rule:**

Two rules, one role:

```
[Region] = "North"           -- rule on RegionDim
[Category] = "Electronics"   -- rule on Sales, same role
```

Read it like English: «Keep only Sales rows that pass the region rule AND the category rule.»

Result: 3 rows (IDs 1,3,9) — **Total Sales 116,000**, Profit 26,500

North alone has 4 rows and Electronics alone has 6; only the 3 rows in both lists survive.

⚠ Common mistake

Creating two separate roles when you want AND. Two roles give the union (next section), which shows *more* data, not less.



10.5 Two roles = union (additive roles)

Why: A person who belongs to several roles sees every row that **any** of the roles allows. Roles add up; they never subtract.

1. Create role **North** (`[Region] = "North"` on RegionDim) and role **West** (`[Region] = "West"` on RegionDim), each in **Modeling** → **Manage roles**.
2. Publish. In the service, add the same person as a member of both roles.
3. Test with **Test as role** and a combination of both roles.

Result: North + West roles: 6 rows (IDs 1,3,4,6,8,9) — **Total Sales 171,000**, Profit 42,500

Result: North + Electronics roles: 7 rows (IDs 1,2,3,4,6,7,9) — **Total Sales 216,000**, Profit 51,500

Compare with **Section 10.4**: as one role, North and Electronics give 3 rows; as two roles they give 7 rows.

	#1 North	#2 South	#3 North	#4 West	#5 South	#6 North	#7 South	#8 West	#9 North	#10 South
Role North	☒	☐	☒	☐	☐	☒	☐	☐	☒	☐
Role Electronics	☒	☒	☒	☒	☐	☐	☒	☐	☒	☐
One role, 2 rules (AND)	☒	☐	☒	☐	☐	☐	☐	☐	☒	☐
Two roles (union)	☒	☒	☒	☒	☐	☒	☒	☐	☒	☐

Figure 10.1 — Intersection inside one role versus union across two roles (cyan = visible)

⚠ Common mistake: a blocking role does not block

“Once denied, always denied” does not apply. A user in a role with `FALSE()` and another role with `TRUE()` sees everything: 10 rows (IDs 1,2,3,4,5,6,7,8,9,10) — **Total Sales 276,000**. Prefer one role that describes each user's full needs.

10.6 Numeric threshold

Why: Junior staff may see only small deals. A comparison on a number column keeps rows under a limit.

Table: Sales · **Editor:** DAX · **Role:** SmallDeals. **Rule:**

```
[SalesAmount] < 50000
```

Read it like English: «Keep a Sales row only when its SalesAmount is below 50,000.»

Result: 8 rows (IDs 2,3,4,5,6,8,9,10) — **Total Sales 126,000**, Profit 40,500

Row 7 has exactly 50,000, so `<` excludes it; use `<=` to include it.

**⚠ Common mistake**

Hoping the role can still show the company total while hiding big deals. RLS cannot hide detail and show summaries: the role sees only totals of the rows it can access, so its Total Sales is 126,000, not 276,000. For a share of the full total see [Section 10.15](#).

10.7 Date rule

Why: A new contractor may see sales only from their start date. Build the date with `DATE` so the comparison is between dates, not text.

Table: Sales · **Editor:** DAX · **Role:** FromJan6. **Rule:**

```
[Date] >= DATE ( 2026, 1, 6 )
```

Read it like English: «Keep a Sales row only when its Date is 6 January 2026 or later.»

Result: 5 rows (IDs 6,7,8,9,10) — **Total Sales 115,000**, Profit 32,000

⚠ Common mistake

Comparing with a text such as "06/01/2026". The Date column must be the Date type ([Chapter 2](#)) and the value must come from `DATE ( year, month, day )`.

10.8 Block all or allow all: FALSE() and TRUE()

Why: Sometimes a role must see nothing from a table, or everything. A rule returns TRUE or FALSE per row; `FALSE()` is false for every row and `TRUE()` is true for every row.

Table: Sales · **Editor:** DAX · **Role:** BlockAll. Repeat for role AllowAll with the second rule. **Rule:**

```
FALSE()  -- role BlockAll, rule on Sales
TRUE()   -- role AllowAll, rule on Sales
```

Read it like English: «BlockAll: no Sales row passes. AllowAll: every Sales row passes.»

Result: BlockAll: (Blank) — no rows

Result: AllowAll: 10 rows (IDs 1,2,3,4,5,6,7,8,9,10) — **Total Sales 276,000**, Profit 70,500

A table with no rule at all is also fully readable, unless a filter reaches it from another table through a relationship. `TRUE()` just makes that intent visible.

⚠ Common mistake

Adding a `FALSE()` role to “take away” access from someone who already has another role. Roles are additive ([Section 10.5](#)), so the other role still wins.



10.9 Dynamic: each salesperson sees their own sales

Why: One role replaces many static ones. `USERPRINCIPALNAME()` returns the signed-in user's UPN, and the rule compares it with the Employee table.

Table: Employee · **Editor:** DAX · **Role:** Salesperson (dynamic rules need the DAX editor). Test: **View as** → the role + **Other user** = `ravi@ima-demo.com`. **Rule:**

```
[Email] = USERPRINCIPALNAME()
```

Read it like English: «Keep the Employee row whose Email equals the signed-in user; Sales follows Employee[Name] → Sales[Salesperson].»

Result: Ravi: 4 rows (IDs 1,3,6,9) — **Total Sales 131,000**, Profit 31,500

Signed-in user	Ravi	Priya	Amit	Meera	Kiran	Unknown
Visible IDs	1,3,6,9	2,5,7,10	4,8	none	none	none
Total Sales	131,000	105,000	40,000	(Blank)	(Blank)	(Blank)
Profit	31,500	28,000	11,000	(Blank)	(Blank)	(Blank)

Meera and Kiran are managers. The rule finds their Employee row, but no Sales row has them as Salesperson, so they see nothing. Managers need either a list of regions ([Section 10.10](#)) or the reporting hierarchy ([Section 10.13](#)).

⚠ Common mistake

Storing the e-mail address instead of the UPN. The UPN is the sign-in name and is not always the same as the mail address; store exactly what `USERPRINCIPALNAME()` returns. Comparisons are case-insensitive, so upper/lower case is not the issue.

10.10 Dynamic: region list through a bridge (bidirectional)

Why: Meera manages North and South. A bridge table, UserRegion, lists every region a person may see, so one person can have many regions and one region many people.

Table: Users · **Editor:** DAX · **Role:** RegionByUser. **Rule:**

```
[UserEmail] = USERPRINCIPALNAME()
```

Read it like English: «Keep my Users row; my UserRegion rows follow; their Regions filter RegionDim; RegionDim filters Sales.»

The filter has to travel from UserRegion (many side) up to RegionDim (one side). By default security filtering is single-direction, even on a bidirectional relationship, so open the RegionDim–UserRegion relationship and set:



Relationship	Cross filter direction	Apply security filter in both directions
Users[UserEmail] 1 → * UserRegion[UserEmail]	Single	Off
RegionDim[Region] 1 → * UserRegion[Region]	Both	On
RegionDim[Region] 1 → * Sales[Region]	Single	Off

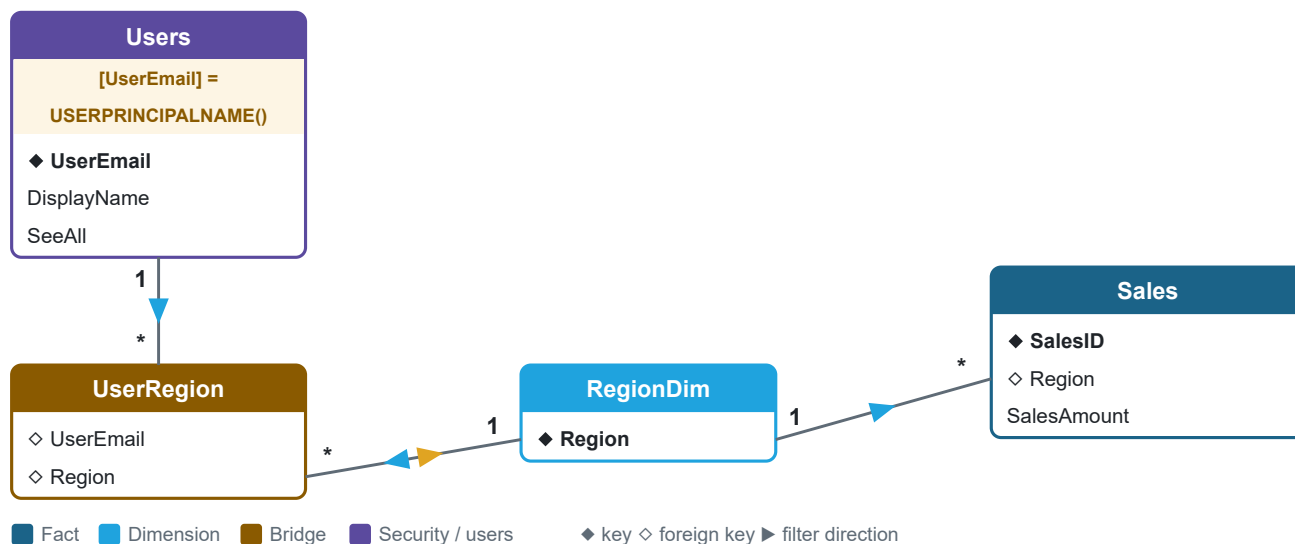


Figure 10.2 — The bridge path: the rule on Users reaches Sales through UserRegion and RegionDim

Signed-in user	Ravi	Priya	Amit	Meera	Kiran	Unknown
Visible IDs	1,3,6,9	2,5,7,10	4,8	1,2,3,5,6,7,9,10	none	none
Total Sales	131,000	105,000	40,000	236,000	(Blank)	(Blank)
Profit	31,500	28,000	11,000	59,500	(Blank)	(Blank)

Result: Meera: 8 rows (IDs 1,2,3,5,6,7,9,10) — **Total Sales 236,000**, Profit 59,500; Kiran: (Blank) — no rows, because she has no UserRegion rows.

⚠ Common mistake

Setting Cross filter direction to Both but forgetting **Apply security filter in both directions**. Then the filter stops at UserRegion and every user sees all regions. If a table takes part in several bidirectional relationships, the option can be selected for only one of them, and it can hurt performance.



10.11 Dynamic: region list with a DAX lookup (no bidirectional)

Why: Same need as the previous section, but all relationships stay single-direction. The rule itself reads the bridge table and returns the user's regions.

Table: RegionDim · **Editor:** DAX · **Role:** RegionByLookup. Leave every relationship single-direction, **Apply security filter in both directions Off**. **Rule:**

```
VAR me = USERPRINCIPALNAME ()
RETURN
    RegionDim[Region]
        IN CALCULATETABLE (
            VALUES ( UserRegion[Region] ), -- regions in the bridge...
            UserRegion[UserEmail] = me      -- ...for the signed-in user only
        )
```

Read it like English: «Keep a RegionDim row when its Region appears in my rows of UserRegion.»

Result: Identical to **Section 10.10** — Meera: 8 rows (IDs 1,2,3,5,6,7,9,10) — **Total Sales 236,000**, Profit 59,500; Ravi: 131,000; Kiran and unknown users: (Blank) — no rows.

Trade-off: the bidirectional version keeps the logic in relationships but bidirectional security filtering can hurt performance; this version needs no relationship change but hides the logic in DAX, and Microsoft advises letting relationships do the work instead of lookups where they can.

Measure both with Performance Analyzer: compare durations without RLS and with **View as**.

⚠ Common mistake

Writing `CALCULATETABLE ( VALUES ( UserRegion[Region] ) )` without the e-mail filter. It returns every region in the bridge, so every user sees North, South and West.



10.12 Dynamic with a See-all flag

Why: Kiran, the director, must see everything without being given every region one by one. The Users table has a **SeeAll** column that is TRUE only for Kiran.

Table: RegionDim · **Editor:** DAX · **Role:** RegionOrAll. No bidirectional setting is needed. **Rule:**

```
VAR me = USERPRINCIPALNAME ()
VAR seeAll =
    LOOKUPVALUE ( Users[SeeAll], Users[UserEmail], me ) -- my flag
RETURN
    seeAll = TRUE ()
        || RegionDim[Region]
            IN CALCULATETABLE (
                VALUES ( UserRegion[Region] ),
                UserRegion[UserEmail] = me
            )
```

Read it like English: «If my SeeAll flag is TRUE keep every region; otherwise keep only my regions from UserRegion.»

Signed-in user	Ravi	Priya	Amit	Meera	Kiran	Unknown
Visible IDs	1,3,6,9	2,5,7,10	4,8	1,2,3,5,6,7,9,10	1,2,3,4,5,6,7,8,9,10	none
Total Sales	131,000	105,000	40,000	236,000	276,000	(Blank)
Profit	31,500	28,000	11,000	59,500	70,500	(Blank)

Total Sales per signed-in user (See-all pattern)

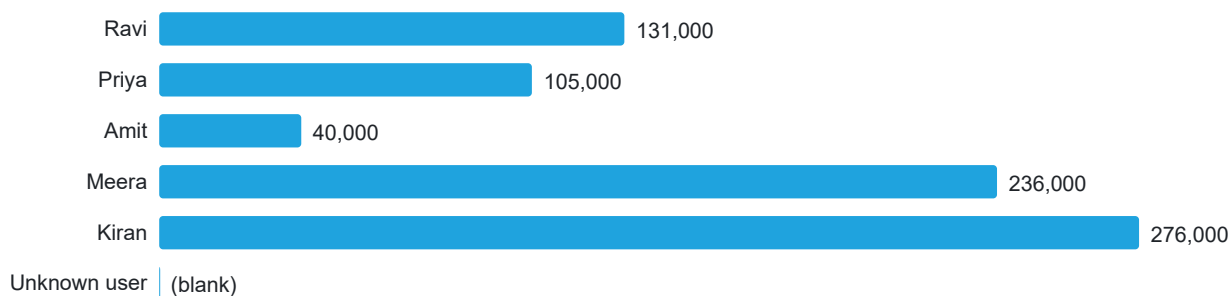


Figure 10.3 — Per-user Total Sales under the See-all rule; the unknown user gets (Blank)

✓ Safe default

For an unknown user `LOOKUPVALUE` finds no row and returns blank, the test is not TRUE, and the bridge has no rows either, so the user sees **(Blank)** — **no rows**. Write rules so that unexpected values return nothing.

**⚠ Common mistake**

Setting SeeAll to TRUE for the wrong person. The flag overrides the bridge, so one wrong cell in Users gives that user every row (276,000). Review the Users table as carefully as the rule.

10.13 Manager hierarchy with PATH and PATHCONTAINS

Why: A manager should see their own team and everyone below them, at any depth. `PATH` stores each employee's chain of managers; `PATHCONTAINS` tests whether the signed-in user is in that chain.

First add a calculated column to Employee; then create the role.

```
Path =
PATH ( Employee[EmployeeID], Employee[ManagerID] )
```

Name	Kiran	Meera	Ravi	Priya	Amit
EmployeeID	1	2	3	4	5
ManagerID	(blank)	1	2	2	1
Path	1	1 2	1 2 3	1 2 4	1 5

Table: Employee · Editor: DAX · Role: Hierarchy. Rule:

```
PATHCONTAINS (
    [Path],
    -- my EmployeeID, found by my sign-in
    LOOKUPVALUE ( Employee[EmployeeID], Employee[Email], USERPRINCIPALNAME () )
)
```

Read it like English: «Keep an employee row when my EmployeeID appears in that employee's chain of managers.»

Signed-in user	Ravi	Priya	Amit	Meera	Kiran	Unknown
Visible IDs	1,3,6,9	2,5,7,10	4,8	1,2,3,5,6,7,9,10	1,2,3,4,5,6,7,8,9,10	none
Total Sales	131,000	105,000	40,000	236,000	276,000	(Blank)
Profit	31,500	28,000	11,000	59,500	70,500	(Blank)

ID 1 (Kiran) is in every path, so she sees all rows. Meera's ID 2 is in her own, Ravi's and Priya's paths; she has no sales of her own, so her total is Ravi's plus Priya's.

⚠ Common mistake

`PATHCONTAINS` is not supported in DirectQuery mode in calculated columns or RLS rules. Use this pattern with an Import Employee table.



10.14 Embedded app passes the identity

Why: In an “app owns data” embedded solution the app, not the viewer, signs in. When the app generates the embed token it passes an **EffectivenessIdentity**: a username, the roles and the datasets. For a dynamic role the username becomes the filter value.

Table: RegionDim · **Editor:** DAX · **Role:** RegionDynamic. **Rule:**

```
RegionDim[Region] = USERNAME ()
```

Read it like English: «Keep the region whose name equals the username the app passed.»

Token request body (one username only; ids are placeholders):

```
{
  "accessLevel": "View",
  "identities": [
    {
      "username": "South",
      "roles": [ "RegionDynamic" ],
      "datasets": [ "<semantic model id>" ]
    }
  ]
}
```

Result: username "South": 4 rows (IDs 2,5,7,10) — **Total Sales 105,000**, Profit 28,000

Result: username "Sout" (typo): **(Blank) — no rows**. This is the safe outcome.

A static role ignores the username, so any string works there. Validate rules with expected and unexpected values, and write them so anything unexpected returns no rows:

Rule that lists the allowed values:

```
VAR v = USERNAME ()
RETURN
    SWITCH (
        TRUE (),
        v = "North", RegionDim[Region] = "North",
        v = "South", RegionDim[Region] = "South",
        v = "West", RegionDim[Region] = "West",
        FALSE () -- anything else: no rows
    )
```

⚠ Common mistake

- Sending no identity. With a service principal, token generation fails; with a master user, the token succeeds but the data is **not filtered**.
- Using `USERPRINCIPALNAME ()` under service-principal embedding: it returns the app ID or an empty string, not the viewer.



CustomData, workspace isolation and other embedding details are in [Chapter 11](#).

10.15 Show a share of the full total

Why: Ravi should see his own sales *and* what percentage of company sales they are. No DAX measure can override RLS or even detect it, so the full total must come from a table that RLS does not filter.

1. Create the calculated table below. Do **not** create any relationship between SalesSummary and RegionDim, Sales or the security tables, so no security filter can reach it.
2. Create the measure below and keep the role from [Section 10.10](#).

Calculated table (unrelated):

```
SalesSummary =
SUMMARIZECOLUMNS ( Sales[Region], "AllSales", SUM ( Sales[SalesAmount] ) )
```

Measure:

```
% of All Sales =
-- secured numerator (my rows) / unsecured denominator (everyone)
DIVIDE ( [Total Sales], SUM ( SalesSummary[AllSales] ) )
```

Read it like English: «My visible sales divided by the sales of every region.»

Signed-in user	Ravi	Priya	Amit	Meera
Total Sales	131,000	105,000	40,000	236,000
% of All Sales	47.46%	38.04%	14.49%	85.51%

⚠ Disclosure risk

Ravi sees 131,000 and 47.46%. Dividing gives the company total 276,000, so he also learns that the other regions sold 145,000 together. With only three regions, Meera (85.51% for North + South) can deduce West exactly: 40,000. Check what a summary reveals before you publish it.

⚠ Common mistake

Relating SalesSummary[Region] to RegionDim. The security filter then reaches the summary and the share becomes 100% for everyone.

10.16 External customer portal

Why: Outside customers see only their own purchases, through a table that maps each guest's sign-in to a Customer.

CustomerUsers (illustrative) maps `amit@partner-a.com` → Amit, `neha@partner-b.com` → Neha. Relate CustomerUsers[Customer] 1 → * Sales[Customer]; it can be the one side because each Customer appears only once in it.

**Table: CustomerUsers · Editor: DAX · Role: Customer. Rule:**

```
[UserEmail] = USERPRINCIPALNAME()
```

Read it like English: «Keep my mapping row; its Customer filters Sales.»

Result: Amit's company: 3 rows (IDs 1,4,8) — **Total Sales 140,000**, Profit 31,000

Result: Neha's company: 2 rows (IDs 2,7) — **Total Sales 55,000**, Profit 12,000

⚠ Common mistake: guest identities

- For B2B guests `USERPRINCIPALNAME()` may return `user@partner.com` or a form like `user_partner.com#EXT#@tenant.onmicrosoft.com`. The format is not guaranteed — check it with a real guest account and store that value.
- Security groups that contain guests might not evaluate correctly; add guests to the role directly by e-mail, or rely on this dynamic rule.

10.17 Summary of all patterns

Pattern	Rule on	DAX rule (UPN = USERPRINCIPALNAME())	User / role	Visible IDs	Total Sales
Static one	RegionDim	<code>[Region] = "North"</code>	North	1,3,6,9	131,000
Static IN	RegionDim	<code>IN {"North","West"}</code>	NorthWest	1,3,4,6,8,9	171,000
Fact column	Sales	<code>[Category]="Electronics"</code>	Electronics	1,2,3,4,7,9	201,000
AND, 1 role	both	North + Electronics	NorthElec.	1,3,9	116,000
2 roles	RegionDim	North; West	North, West	1,3,4,6,8,9	171,000
Threshold	Sales	<code>[SalesAmount]<50000</code>	SmallDeals	2,3,4,5,6,8,9,10	126,000
Date	Sales	<code>[Date]>=DATE(2026,1,6)</code>	FromJan6	6,7,8,9,10	115,000
Block	Sales	<code>FALSE()</code>	BlockAll	none	(Blank)
Allow	Sales	<code>TRUE()</code>	AllowAll	1,2,3,4,5,6,7,8,9,10	276,000
Own sales	Employee	<code>[Email] = UPN</code>	Ravi	1,3,6,9	131,000
Bridge Both	Users	<code>[UserEmail] = UPN</code>	Meera	1,2,3,5,6,7,9,10	236,000
Bridge DAX	RegionDim	<code>IN CALCULATETABLE(...)</code>	Meera	1,2,3,5,6,7,9,10	236,000
See-all flag	RegionDim	<code>seeAll = TRUE() ...</code>	Kiran	1,2,3,4,5,6,7,8,9,10	276,000
Hierarchy	Employee	<code>PATHCONTAINS([Path], ...)</code>	Meera	1,2,3,5,6,7,9,10	236,000
Embedded	RegionDim	<code>[Region] = USERNAME()</code>	"South"	2,5,7,10	105,000
% of all	(measure)	<code>DIVIDE(...)</code>	Ravi	1,3,6,9	47.46%
Portal	CustomerUsers	<code>[UserEmail] = UPN</code>	Amit (cust.)	1,4,8	140,000



11. RLS in Special Scenarios

Storage mode means where the data sits when a visual asks a question: copied into the model (**Import**), queried live from the source (**DirectQuery**) or read from Fabric OneLake (**Direct Lake**). With a **live connection** the report has no model of its own.

Mode	Where RLS is defined	Who enforces it	Testing
Import	Model roles. Security roles in the source are not used.	Power BI model	View as, Test as role
DirectQuery	Model roles + source security with the user's credentials. Tune the source (indexes, persisted computed columns).	Model + source	View as, Test as role
DirectQuery with SSO	Source RLS (e.g. Azure SQL) can be used.	Source (+ model roles)	View as / Test as role don't work ; use real accounts
Direct Lake	Model roles; still applied on DirectQuery fallback (speed may change).	Power BI model	View as, Test as role
Live connection to AS / Azure AS	In the AS model; Desktop has no roles option.	Analysis Services	In the AS model
Composite model with a remote model	On the source model. You can't define RLS on remote tables, local roles don't filter them, and RLS isn't imported. A DirectQuery Users table shows new users after the next refresh.	Source model	On the source model
Power BI Report Server	Roles in Desktop; members (DOMAIN\user) in Manage → Row-level security , by Publisher or Content Manager only.	Report Server	Not respected for DirectQuery with integrated authentication

Model-creation limits: RLS needs a model created with Power BI Desktop (convert Excel models to .pbix); roles from the old service experience must be re-created in Desktop (**Chapter 5**).

11.1 Embedded analytics: app owns data

Your own web application shows Power BI reports to users who never sign in to Power BI. The app signs in as a **master user** (a real Power BI account) or a **service principal** (an app identity in Microsoft Entra), so the app must tell Power BI whose data to show.



Figure 11.1 — How an embedded app passes an identity to Power BI

What it is:

EffectivenessIdentity is the identity block sent when the embed token is generated: **username** (only one), **roles** and **datasets** (the semantic models). Optional **CustomData** is read in a rule with CUSTOMDATA() (blank if not set; not supported in RLS rules on DirectQuery models). A **static** role ignores the username; a **dynamic** role uses it as the filter value.



Syntax / where to click:

```
-- Role RegionDynamic, rule on RegionDim (DAX editor)
RegionDim[Region] = USERNAME ()
```

The app then sends an identity with `"username": "South", "roles": [ "RegionDynamic" ]` and the semantic model id in `"datasets"` (full request body in [Section 10.14](#)).

Simple example:

The app sends username `South` with role `RegionDynamic` ([Section 10.14](#)). `USERNAME()` returns the value from the token.

Result: Rows 2,5,7,10 (4 rows): Total Sales = 105,000, Total Cost = 77,000, Profit = 28,000

Why this result:

Only `RegionDim[South]` passes, and it filters Sales. Role assignments in the service play no part: the token identity applies even to workspace Admins, Members and Contributors.

⚠ Common mistake

- A typo such as `Sout` matches no region: (Blank) — no rows, yet the token is created, so the report just looks empty.
- Service principal without `EffectivelyIdentity`; expecting local composite roles to filter remote tables.
- Testing SSO, guest or embedded rules only with View as / Test as role.

Situation	Master user	Service principal
Token without identity/role on an RLS model	Succeeds, data not filtered	Generation fails
<code>USERPRINCIPALNAME()</code> / <code>USERNAME()</code> in the rule	Pass the username you want the rule to see	Return the app ID or an empty string — not the viewer
Can be added to an RLS role in the service?	Yes, but with an embed token the token identity decides	No — RLS isn't applied when it is the final identity

Many customers? **Workspace-based isolation** is the alternative ([Chapter 13](#)). Token-based identity (IdentityBlob) works only for DirectQuery to Azure SQL.

11.2 Direct Lake and Fabric Warehouse

A Fabric **Warehouse** can have its own T-SQL RLS: an inline table-valued function is the predicate and `CREATE SECURITY POLICY` attaches it. Direct Lake queries on such a warehouse fall back to DirectQuery so the warehouse rules are respected.

Expected behaviour: Warehouse RLS is enforced for report viewers; queries run as DirectQuery, so performance characteristics can change.

External guests (B2B): guests can be role members. Their UPN format (e.g. `user_partner.com#EXT#@tenant.onmicrosoft.com`) is not guaranteed. Security groups containing guests might not evaluate correctly: add guests directly by email or use dynamic RLS. **Test as role** uses **your** identity, so validate with real guest accounts.



12. Object-Level Security (OLS) Companion

What it is:

RLS filters rows only; a user who sees a row sees all its columns. **Object-level security (OLS)** hides whole tables or columns from report viewers. It lives in the same model roles, in the property **metadataPermission**: `none` = hidden, `read` = visible.

Syntax / where to click:

1. Open **TMDL view** (generally available in Desktop; on the web **in preview at the time of writing**) and a new script tab.
2. Type a `createOrReplace` script (below).
3. **Preview** to review the changes, then **Apply**.
4. Publish, then assign members on the **Security** page.
5. Alternative: **External tools** → **Tabular Editor** → Model view → Roles → role → Table Permissions → None/Read; save; publish; assign members.

Simple example:

Hide Sales[Cost] (column) and Employee (whole table)

```
createOrReplace
    role 'Sales Reps'
        modelPermission: read
        tablePermission Sales
            columnPermission Cost
                metadataPermission: none

    role 'No HR'
        modelPermission: read
        tablePermission Employee
            metadataPermission: none
```

⚠ Common mistake

Deleting the Cost visuals instead of securing the column — the column is still reachable.

Visual uses...	Viewer in 'Sales Reps'	Admin / Member / Contributor
[Total Sales], [Total Quantity]	Works (Total Sales = 276,000 with no RLS)	Works
Sales[Cost], [Total Cost], [Profit]	Restricted automatically (they reference Cost): same message as a deleted field	Works

Expected behaviour: Give audiences without Cost their own pages or report, so nobody meets broken visuals.

Restrictions: OLS applies to Viewers only (Admin, Member and Contributor see everything); it is not supported with Quick insights, Smart narrative or the Excel Data Types gallery; RLS and OLS coming from **different roles** give a query-time error, so keep them in one role; and you can't secure a table that breaks a relationship chain.



13. Alternatives and When Not to Use RLS

Option	Restricts	Defined in	Applies to	Pros	Cons / limits
RLS	Rows	Model roles	Viewers	One model for all audiences	No Publish to web
OLS	Tables, columns	Model roles	Viewers	Hides objects	Visuals break; same role as RLS
Separate models per audience	Whole model	Own workspace each (parameters)	Workspace users	Fast, small; Publish to web	Many workspaces, duplicated content, no consolidated view
Workspace isolation (embedded)	Workspace per customer	Workspaces	Embedded app's customers	Recommended with service principal profiles	One workspace per customer
Source security	Source rows	Warehouse policy; SSO	Identities at source	Reuses source rules	DirectQuery performance; SSO models can't use View as / Test as role
OneLake security	Rows, columns, tables	OneLake roles	Not Admin, Member, Contributor	Defined on the data	Third-party engines in preview at the time of writing
Report filters / slicers	Nothing — not security	Report	—	Easy	RLS/OLS don't apply to filter and slicer selections

Workspace / app access

who can open the content at all

Semantic model permissions

Read, Build, Write — Write skips RLS and OLS

RLS and OLS in model roles

rows, tables and columns per Viewer

Source security

Warehouse RLS, DirectQuery SSO, OneLake security

Figure 13.1 — Security layers from outside to inside

13.1 Decision guide

1. Only a convenience, not a must? A filter is fine. Admin, Member or Contributor? RLS/OLS don't apply — use Viewer or an app.
2. Few static audiences? Separate models. Embedded, many customers? Consider workspace isolation.
3. Columns or tables to hide? Add OLS in the same role. Secured at the source? Reuse it. Otherwise: dynamic RLS with a security table ([Chapter 8](#)).

Good practice: map security groups to roles · give consumers the Viewer role or an app · one role per audience (roles add up) · test every role with expected and unexpected values.



14. Common Mistakes and Troubleshooting

Most RLS problems fall into a few families: the wrong person is tested, the filter cannot travel to the fact table, the user name in the mapping table does not match, or the user is not filtered at all because of their workspace role. This chapter lists the symptoms you will actually see and how to fix each one. Numbers come from the practice model of [Chapter 2](#).

14.1 Symptom → cause → fix

Symptom	Likely cause	Fix
A user sees everything although a role exists.	The user is a workspace Admin , Member or Contributor (or has Write on the model). RLS filters only users with Viewer/Read permission.	Give report readers the Viewer role, or share an app or the report instead. Chapter 7
A user sees nothing : every card shows (Blank).	The user is not a member of any role. With RLS defined, Read/Build users outside every role see no data.	Add the user (better: their security group) to the role on the Security page.
Dynamic rule: one person sees (Blank), others are fine.	The value in the mapping table does not match what USERPRINCIPALNAME() returns — typo, a changed name, or the email stored instead of the UPN.	Show the Who Am I card for that user and copy the exact text into the mapping table.
A guest (B2B) user sees nothing.	For guests USERPRINCIPALNAME() may return <code>user@partner.com</code> or a <code>#EXT#</code> form; the format is not guaranteed. Security groups with guests may not evaluate correctly.	Check the real value with a guest account. Add guests directly by email, or use dynamic RLS.
You cannot add a group to the role.	It is a Microsoft 365 group or a group created in Power BI — both are not supported.	Use a Microsoft Entra security group, a distribution group or a mail-enabled group.
A user in two roles sees more than expected.	Roles are additive: the visible rows are the union of all roles. A FALSE() role does not cancel a TRUE() role.	Give each user one role that matches their needs. Section 10.5
Meera sees all 276,000 — or nothing — with the bridge design.	The UserRegion → RegionDim relationship does not filter in both directions for security, so the filter on Users stops at UserRegion.	Set cross filter direction to Both and tick Apply security filter in both directions , or use the DAX lookup. Section 10.10
The rule works on RegionDim but Sales is not filtered.	The relationship is inactive (dotted line). Security filters travel only through active relationships.	Make the relationship active, or write the rule on the fact table.
A measure using USERRELATIONSHIP() throws an error after RLS is added.	With RLS enabled, USERRELATIONSHIP() may cause unexpected errors.	Model the second relationship differently (for example a role-playing copy of the dimension).
Test as role shows your own data for a dynamic role.	Test as role evaluates USERPRINCIPALNAME() with your identity.	Use Now viewing as → a specific person, then confirm with real accounts. Chapter 6
View as / Test as role is not available.	The model is DirectQuery with single sign-on (SSO) enabled.	Test with real user accounts.



Symptom	Likely cause	Fix
A Q&A visual, Quick insights, Copilot, a dashboard or a paginated report cannot be tested.	Test as role does not cover these items; dashboards redirect to the report.	Test the underlying report, then check the rest with a real Viewer account.
A DirectQuery model rejects the rule.	CUSTOMDATA() and PATHCONTAINS() are not supported in DirectQuery mode inside RLS rules or calculated columns.	Use Import for the security tables, or rewrite the rule with simple comparisons.
“Security” does not help: a slicer still shows a hidden region.	RLS does not apply to values stored outside the model, such as saved slicer and filter selections.	Never use a slicer or report filter as security; keep the restriction in a role.
A visual shows a “field not found”-style error for some users.	OLS hides that column or table; the message looks like a deleted field.	Design separate visuals or pages for secured objects. Chapter 12
Embedded report shows all data to every customer.	The embed token was generated with a master user and no identity — the token works but data is not filtered.	Always pass an EffectiveIdentity with username and roles. Section 10.14
The Security page has no roles for an older model.	The model was created in Excel, or roles were created in the old service experience.	Convert the Excel model to a PBIX; re-create the roles in Power BI Desktop.
Totals differ between visuals for the same user.	Aggregation tables are not filtered in the same way as the detail table, or a table is empty.	Apply the same security to aggregation tables; check every table has data.

⚠ Common mistake: “fixing” it with a measure

A DAX measure cannot override RLS and cannot even detect that RLS is active. If numbers are wrong, the cause is in the roles, the relationships or the mapping data — never patch it in a measure.

14.2 A six-step debug routine

Work through the steps in order. Each step rules out one family of problems, so you never guess.

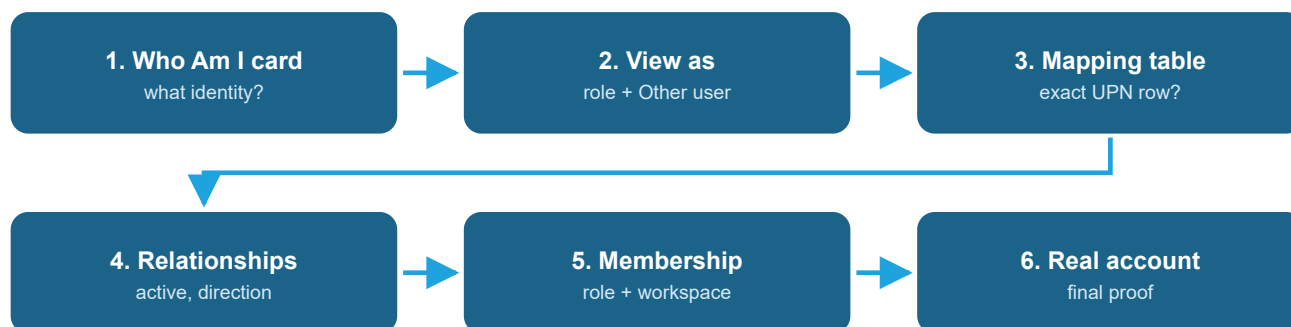


Figure 14.1 — The RLS debug routine: identity first, real account last

1. **Who Am I card.** Add a measure and put it on a card. It shows exactly which text the rule compares.

```

Who Am I =
-- shows the identity the rule will compare
USERPRINCIPALNAME ()
  
```



2. **View as.** In Desktop: **Modeling** → **View as** → tick the role → tick **Other user** and type the UPN → **OK**.
Compare the totals with your expected table.
3. **Check the mapping table.** Is there a row for that UPN? Is it the UPN, not the mail address? (Case does not matter: DAX text comparison is case-insensitive.)
4. **Check relationships.** In the model view: is every relationship on the path active? Does the filter need to travel against the arrow (then you need **Apply security filter in both directions** or a DAX lookup)?
5. **Check membership.** In the service: semantic model **More options (...)** → **Security**: is the user or their security group in the role? In the workspace: is the user a **Viewer**?
6. **Test with the real account.** **Test as role** uses your identity; the final proof is a real Viewer opening the report.

Expected behaviour: After step 6 each test user sees exactly the numbers in the acceptance table of **Chapter 15**; a user with no mapping row sees (Blank) — no rows.



15. End-to-End Mini Project

Build the **IMA Sales** app from an empty file to a tested, shared report; tick each box as you go.

15.1 The brief

Who	UPN (fictitious)	Must see
Ravi	ravi@ima-demo.com	North only
Priya	priya@ima-demo.com	South only
Amit	amit@ima-demo.com	West only
Meera (manager)	meera@ima-demo.com	North and South
Kiran (director)	kiran@ima-demo.com	Everything (SeeAll = TRUE)
Anyone else	—	Nothing

Readers must be **Viewers**, never editors. One role, **Sales Security**, serves everybody: dynamic RLS with a See-all flag ([Section 10.12](#)), using a DAX lookup so no relationship needs to filter both ways.

15.2 Build checklist

- ☐ Load **Sales**, **RegionDim**, **Users** and **UserRegion** as in [Chapter 2](#); create [Total Sales], [Total Cost], [Profit], [Total Quantity] and a **Who Am I** card.
- ☐ Relationships, all **active** and single direction: RegionDim → Sales, Users → UserRegion, RegionDim → UserRegion.
- ☐ **Modeling** → **Manage roles** → **New** → **Sales Security** → **Select tables** → RegionDim → **Switch to DAX editor** → type the rule below → checkmark → **Save**.

```
-- Role: Sales Security · table: RegionDim
VAR CurrentUser = USERPRINCIPALNAME ()
VAR UserSeesAll =
    LOOKUPVALUE ( Users[SeeAll], Users[UserEmail], CurrentUser )
VAR UserRegions =
    CALCULATETABLE (
        VALUES ( UserRegion[Region] ),
        UserRegion[UserEmail] = CurrentUser
    )
RETURN
    -- unknown users: no flag and no regions -> no rows
    UserSeesAll = TRUE () || RegionDim[Region] IN UserRegions
```

Read it like English:

«If my SeeAll flag is TRUE keep every region; otherwise keep only the regions mapped to me.»

- ☐ **Modeling** → **View as** → **Sales Security + Other user** → each UPN (and stranger@ima-demo.com) → **OK**; compare with the acceptance table.



- ☐ Publish → semantic model **More options (...)** → **Security** → Sales Security → add the **security group** of the five users (not a Microsoft 365 group).
- ☐ Give readers the workspace **Viewer** role (or share an app).
- ☐ **More options (...)** → **Test as role** → **Now viewing as** → each person → **Back to Row-Level Security**; finally ask one real Viewer to read the Who Am I card.

15.3 Acceptance test

Every number below is what the report must show for that user.

Test user	Total Sales	Profit	Total Quantity	Rows (SalesIDs)
Ravi	131,000	31,500	16	4 (1,3,6,9)
Priya	105,000	28,000	12	4 (2,5,7,10)
Amit	40,000	11,000	4	2 (4,8)
Meera	236,000	59,500	28	8 (1,2,3,5,6,7,9,10)
Kiran	276,000	70,500	32	10 (1,2,3,4,5,6,7,8,9,10)
stranger@ima-demo.com	(Blank) — no rows	(Blank)	(Blank)	0
Any workspace Member (not filtered)	276,000	70,500	32	10 (all)

Result: Meera = North 131,000 + South 105,000 = 236,000; Kiran = all 276,000; unknown users = (Blank) — no rows.

⚠ Why the Member row shows everything

RLS filters only Viewer (Read) users. Admins, Members and Contributors have Write access and are never filtered — always test with **Test as role** or a real Viewer.



16. Practice Exercises

Use the data from [Section 2.1](#) and [Section 2.4](#). Answers are in [Chapter 17](#). No rows means (Blank), not 0. Prove each result with **Modeling** → **View as**.

16.1 Beginner (exercises 1–5)

#	Type	Exercise
1	Write the rule	Write a static rule on Sales so a role sees only the South region. Total Sales?
2	Predict the total	Rule <code>[Category] = "Furniture"</code> on Sales. Total Sales and Profit?
3	Which rows survive	Rule <code>[Region] <> "North"</code> . Which SalesIDs survive?
4	Predict the total	A role contains only <code>FALSE()</code> on Sales. What does a [Total Sales] card show?
5	Find the bug	You try to save a role called <code>North,West</code> . Power BI refuses. Why?

16.2 Intermediate (exercises 6–10)

#	Type	Exercise
6	Write the rule	One role must show only Furniture sold in the South . Write one rule; give the total.
7	Predict the total	Neha is in two roles: South (<code>[Region] = "South"</code>) and Electronics (<code>[Category] = "Electronics"</code>). What does she see?
8	Write the rule	Keep only order lines with a quantity of 3 or more . Total Sales and Total Quantity?
9	Predict the total	A customer portal role uses <code>[Customer] = "Rahul"</code> . What does Rahul see?
10	Which users are filtered	Workspace Admin, Member, Contributor, Viewer and a Viewer with Build all belong to the role <code>[Region] = "West"</code> . Who sees only West?

16.3 Advanced (exercises 11–15)

#	Type	Exercise
11	Find the bug	Bridge design (Users → UserRegion ← RegionDim), rule <code>[UserEmail] = USERPRINCIPALNAME()</code> on Users, no relationship has Apply security filter in both directions . What does Ravi see, and why?
12	Predict the total	Same bridge design, fixed. Total Sales for Meera and for Kiran (no UserRegion rows; the rule ignores SeeAll)?
13	Find the bug	The mapping table stores <code>ravi.k@ima-demo.com</code> but Ravi signs in as <code>ravi@ima-demo.com</code> . Rule: <code>Employee[Email] = USERPRINCIPALNAME()</code> . What does Ravi see?
14	Write the rule	Write the hierarchy rule on Employee so a manager sees their whole team. Predict Meera's result.
15	Predict the total	Priya's report divides her sales by all company sales from an unsecured summary table. What percentage does she see, and what can she deduce?



17. Answer Key

Rule (where relevant) or key point, then the exact result from the sample data.

#	Rule / key point	Result
Beginner		
1	[Region] = "South"	Total Sales = 105,000 · Profit = 28,000 · Total Quantity = 12 · 4 rows (IDs 2,5,7,10)
2	—	Total Sales = 75,000 · Profit = 24,000 · Total Quantity = 9 · 4 rows (IDs 5,6,8,10)
3	—	Total Sales = 145,000 · Profit = 39,000 · Total Quantity = 16 · 6 rows (IDs 2,4,5,7,8,10)
4	FALSE() keeps no row.	(Blank) — no rows
5	Role names cannot contain a comma; rename it, e.g. NorthWest .	The role saves.
Intermediate		
6	[Category] = "Furniture" && [Region] = "South" (one rule = AND)	Total Sales = 50,000 · Profit = 16,000 · Total Quantity = 6 · 2 rows (IDs 5,10)
7	Roles are additive (union).	Total Sales = 251,000 · Profit = 62,500 · Total Quantity = 29 · 8 rows (IDs 1,2,3,4,5,7,9,10)
8	[Quantity] >= 3	Total Sales = 41,000 · Profit = 14,500 · Total Quantity = 22 · 4 rows (IDs 2,3,5,9)
9	—	Total Sales = 31,000 · Profit = 11,500 · Total Quantity = 14 · 3 rows (IDs 3,6,9)
10	Only the Viewer and the Viewer with Build; Write access is not filtered.	Viewers: Total Sales = 40,000 · Profit = 11,000 · Total Quantity = 4 · 2 rows; Admin/Member/Contributor: all 276,000.
Advanced		
11	The filter stops at UserRegion and never reaches RegionDim, so Sales is unfiltered.	Everything: 276,000. After the fix: 131,000.
12	—	Meera: Total Sales = 236,000 · Profit = 59,500 · Total Quantity = 28 · 8 rows (IDs 1,2,3,5,6,7,9,10). Kiran: (Blank) — no rows.
13	No Employee row matches his UPN; store the UPN.	Before the fix: (Blank) — no rows. After: 131,000.
14	PATHCONTAINS (Employee[Path], LOOKUPVALUE (Employee[EmployeeID], Employee[Email], USERPRINCIPALNAME ()))	Total Sales = 236,000 · Profit = 59,500 · Total Quantity = 28 · 8 rows (IDs 1,2,3,5,6,7,9,10)
15	105,000 / 276,000; she can deduce the other regions' combined total (disclosure risk).	38.04%



18. Cheat Sheet and Sources

18.1 One-page reference

Task	Where to click / DAX
Create a role	Modeling → Manage roles → New → name → Select tables → Filter data → Save; dynamic rules need Switch to DAX editor (commas as separators)
Static rules	<code>[Region] = "North" · [Region] IN { "North", "West" } · FALSE() / TRUE()</code>
Per-user rule	<code>Employee[Email] = USERPRINCIPALNAME()</code>
Manager sees team	<code>PATHCONTAINS(Employee[Path], <my EmployeeID>)</code>
Test in Desktop	Modeling → View as → role (+ Other user) → OK
Add members	Semantic model More options (...) → Security → role → add user or group
Test in the service	More options (...) → Test as role → Now viewing as
Bridge filter	Cross filter direction Both + Apply security filter in both directions
Cost / who is filtered	Performance Analyzer (without RLS vs View as) · Viewers incl. Build are filtered; Admin, Member, Contributor are not

18.2 Which approach?

Requirement	Approach	Read
A few fixed audiences	Static role per audience	Section 10.1
Each person sees own rows	Dynamic rule with USERPRINCIPALNAME()	Section 10.9
People see several regions	Mapping (bridge) table	Section 10.11
Some people see everything	SeeAll flag in Users	Section 10.12
Managers see their team	PATH / PATHCONTAINS	Section 10.13
Show share of company total	Unsecured summary table	Section 10.15
App owns data (embedded)	EffectivelyIdentity in embed token	Section 10.14
Hide a column or table	Object-level security	Chapter 12
Public website / tiny static split	Separate models, no RLS	Chapter 13

18.3 Identity functions

Function	Returns	Watch out
<code>USERPRINCIPALNAME()</code>	UPN of the signed-in user	UPN may differ from email; guests vary
<code>USERNAME()</code>	DOMAIN\User in Desktop; UPN in the service	Different in Desktop and service
<code>USEROBJECTID()</code>	Microsoft Entra Object ID	SID in SSAS
<code>CUSTOMDATA()</code>	CustomData of the connection; blank if not set	Not in DirectQuery RLS rules



18.4 Sources

Facts checked on Microsoft Learn on 3 October 2026. All explanations are in our own words; all numbers are computed from the IMA sample data.

Key	Page	URL (accessed 3 October 2026)
S01	Row-level security (RLS) with Power BI	https://learn.microsoft.com/en-us/fabric/security/service-admin-row-level-security
S02	RLS guidance in Power BI Desktop	https://learn.microsoft.com/en-us/power-bi/guidance/rls-guidance
S03	Object-level security (OLS) with Power BI	https://learn.microsoft.com/en-us/fabric/security/service-admin-object-level-security
S04	Standard cloud RLS with embedded content	https://learn.microsoft.com/en-us/power-bi/developer/embedded/cloud-rls
S05	Security in Power BI embedded analytics	https://learn.microsoft.com/en-us/power-bi/developer/embedded/embedded-row-level-security
S06	Bidirectional cross-filtering in Power BI Desktop	https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-bidirectional-filtering
S07	RLS in Power BI Report Server (contains the Desktop "View as" steps)	https://learn.microsoft.com/en-us/power-bi/report-server/row-level-security-report-server
S08	Edit semantic models in the Power BI service	https://learn.microsoft.com/en-us/power-bi/transform-model/service-edit-data-models
S09	Implementation planning: Report consumer security planning	https://learn.microsoft.com/en-us/power-bi/guidance/powerbi-implementation-planning-security-report-consumer-planning
S10	Roles in workspaces in Power BI	https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-roles-new-workspaces
S11	USERPRINCIPALNAME function (DAX)	https://learn.microsoft.com/en-us/dax/userprincipalname-function-dax
S12	USERNAME function (DAX)	https://learn.microsoft.com/en-us/dax/username-function-dax
S13	USEROBJECTID function (DAX)	https://learn.microsoft.com/en-us/dax/userobjectid-function-dax
S14	CUSTOMDATA function (DAX)	https://learn.microsoft.com/en-us/dax/customdata-function-dax
S15	Generate an embed token	https://learn.microsoft.com/en-us/power-bi/developer/embedded/generate-embed-token
S16	Publish to web from Power BI	https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-publish-to-web
S17	Semantic model permissions	https://learn.microsoft.com/en-us/power-bi/connect-data/service-datasets-permissions
S18	Use composite models in Power BI Desktop	https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-composite-models
S19	Object-level security (Analysis Services tabular) — Restrictions	https://learn.microsoft.com/en-us/analysis-services/tabular-models/object-level-security
S20	TMDL view in Power BI Desktop	https://learn.microsoft.com/en-us/power-bi/transform-model/desktop-tmdl-view
S21	Row-Level Security in Fabric Data Warehouse	https://learn.microsoft.com/en-us/fabric/data-warehouse/row-level-security
S22	OneLake table, column, and row-level security	https://learn.microsoft.com/en-us/fabric/onelake/security/table-column-row-security
S23	Performance Analyzer (referenced by S02 for measuring RLS impact)	https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-performance-analyzer
S24	Roles in tabular models (Analysis Services) — Row filters, Dynamic security	https://learn.microsoft.com/en-us/analysis-services/tabular-models/roles-ssas-tabular
S25	PATHCONTAINS function (DAX)	https://learn.microsoft.com/en-us/dax/pathcontains-function-dax
S26	PATH function (DAX)	https://learn.microsoft.com/en-us/dax/path-function-dax
S27	LOOKUPVALUE function (DAX)	https://learn.microsoft.com/en-us/dax/lookupvalue-function-dax